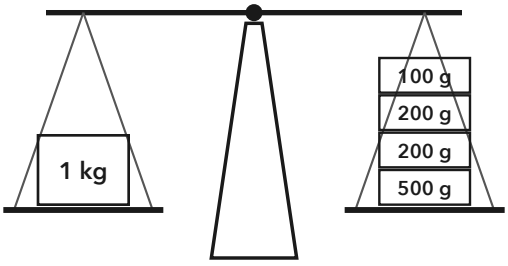
# Mass and Liquid Volume

## How heavy is it? How much does it hold?

### Grams for light things, kilograms for heavy things

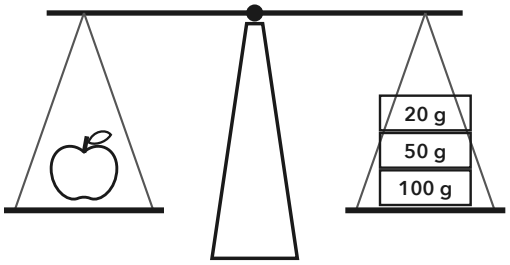
A paper clip is about 1 gram (g).




**1 kilogram = 1000 grams**

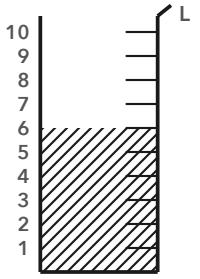
The pans are level: they weigh the same.

### Read the balance: add the weights




**100 + 50 + 20 = 170 g**

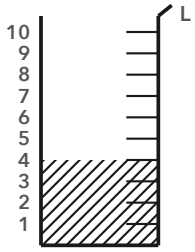
### Read the beaker: count the marks




**6 liters (L)**

Each mark is 1 liter.

### Your turn!




**4** liters

**For teachers**

Goal: measure mass in grams and kilograms and liquid volume in liters, and solve problems with them (3.MD.2). Hold a paper clip (1 g) and a big book (about 1 kg). Say "Is it light or heavy? Grams or kilograms?"

# Mass and Liquid Volume: Practice

Grams, kilograms and liters.

**Part 1** Grams or kilograms? Light things are grams. Heavy things are kilograms.

1) **A grape is about 5** (grams) kilograms

2) **A big dog is about 30** grams (kilograms)

3) **A paper clip is about 1** (grams) kilograms

4) **A bike is about 12** grams (kilograms)

5) **A strawberry is about 15** (grams) kilograms

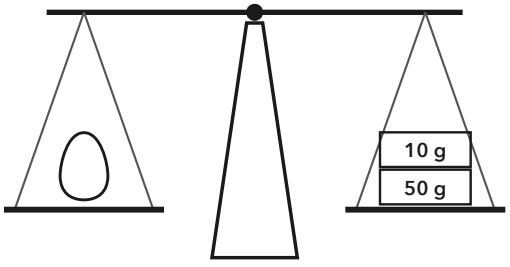
6) **A bag of rice is about 2** grams (kilograms)

7) **A pencil is about 6** (grams) kilograms

8) **A watermelon is about 5** grams (kilograms)

**Part 2** Read the balance Add the weights on the other pan.

9)

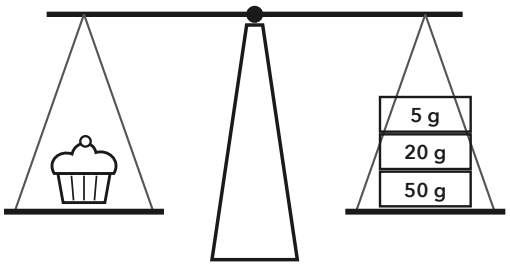



**50 + 10 =** [60] **g**

## Mass and Liquid Volume: Practice (continued)

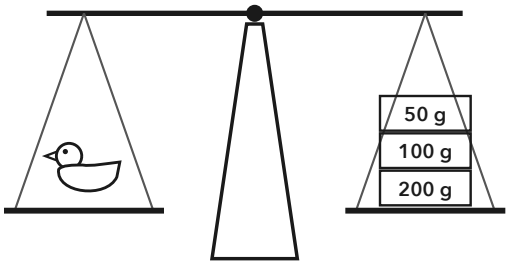
**Part 2** Read the balance (continued)

10)



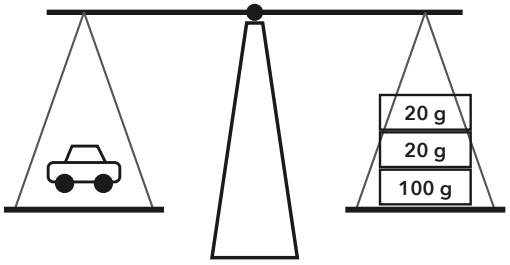

50 + 20 + 5 = 75 g

11)



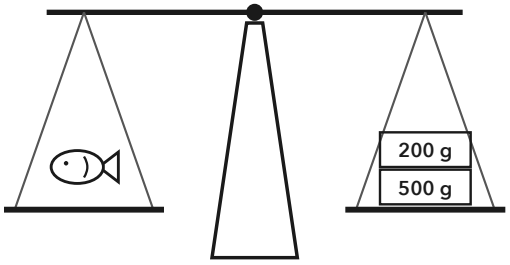

200 + 100 + 50 = 350 g

12)




100 + 20 + 20 = 140 g

13)

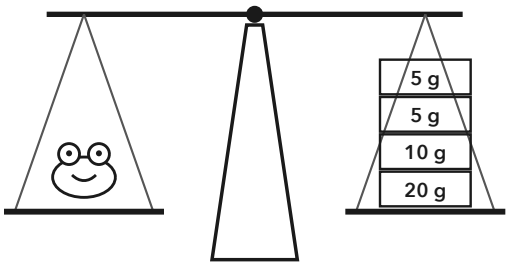



500 + 200 = 700 g

## Mass and Liquid Volume: Practice (continued)

### Part 2 Read the balance (continued)

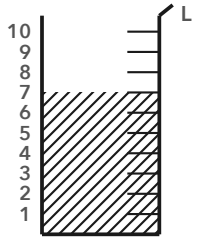
14)




20 + 10 + 5 + 5 = 40 g

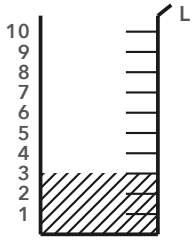
### Part 3 Read the beaker Find the top of the water. Count the marks.

15)



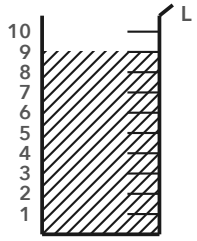

7 liters

16)



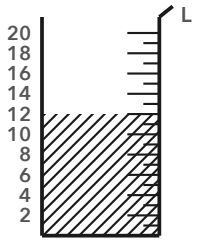

3 liters

17)



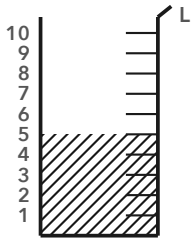

9 liters

18)



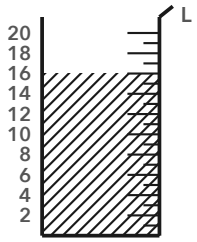

12 liters

19)




5 liters

20)

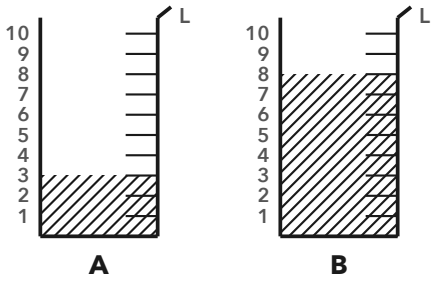



16 liters

## Mass and Liquid Volume: Practice (continued)

**Part 4** How much more? Take away the smaller amount.

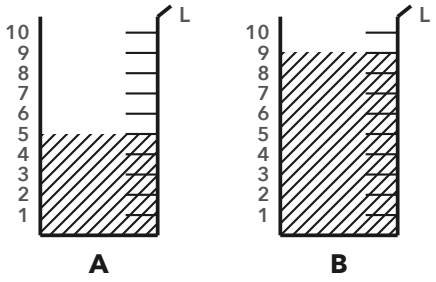
21)




How much more is in B?

8 L − 3 L = 5 L

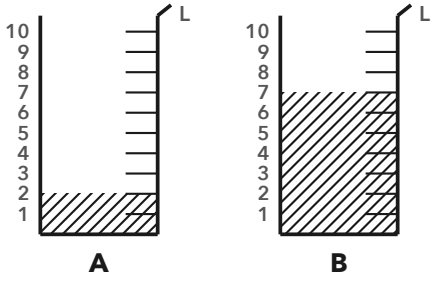
22)




How much more is in B?

9 L − 5 L = 4 L

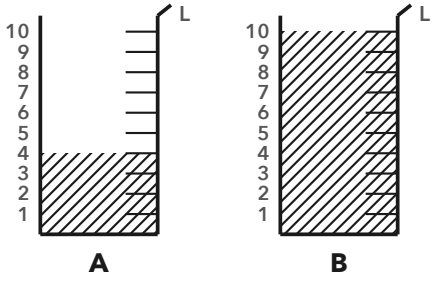
23)




How much more is in B?

7 L − 2 L = 5 L

24)




How much more is in B?

10 L − 4 L = 6 L

## Mass and Liquid Volume: Practice (continued)

**Part 5** Add and take away grams No help this time.

25) 350 g + 240 g = 590 g

26) 800 g − 350 g = 450 g

27) 125 g + 475 g = 600 g

28) 1000 g − 400 g = 600 g

29) 460 g + 270 g = 730 g

30) 900 g − 625 g = 275 g

31) 3 kg + 6 kg = 9 kg

32) 15 kg − 8 kg = 7 kg

**Part 6** Liters No help this time.

33) 4 bottles of 2 L = 8 L

34) 3 jugs of 5 L = 15 L

35) 12 L in 3 equal jugs = 4 L each

36) 7 L + 8 L = 15 L

## Mass and Liquid Volume: Practice (continued)

### Part 6 Liters (continued)

37) 20 L − 6 L = 14 L

38) 6 buckets of 4 L = 24 L

### Part 7 Add and take away

39) 594 mL − 247 mL = 347 mL

40) 656 kg + 145 kg = 801 kg

41) 406 kg + 334 kg = 740 kg

42) 457 mL − 328 mL = 129 mL

43) 661 kg + 328 kg = 989 kg

44) 503 mL − 368 mL = 135 mL

45) 316 mL − 212 mL = 104 mL

46) 299 mL − 52 mL = 247 mL

47) 640 L − 330 L = 310 L

48) 760 kg − 153 kg = 607 kg

## Mass and Liquid Volume: Practice (continued)

**Part 7** Add and take away (continued)

49) 694 kg − 167 kg = 527 kg

50) 389 g + 25 g = 414 g

51) 87 g + 31 g = 118 g

52) 277 g + 200 g = 477 g

53) 561 L − 332 L = 229 L

54) 650 g − 264 g = 386 g

55) 782 L + 75 L = 857 L

56) 455 g + 353 g = 808 g

57) 452 g − 21 g = 431 g

58) 626 kg + 70 kg = 696 kg

59) 317 mL − 92 mL = 225 mL

60) 576 g + 328 g = 904 g

## Mass and Liquid Volume: Practice (continued)

**Part 7** Add and take away (continued)

61) 862 g − 123 g = 739 g

62) 746 g + 225 g = 971 g

63) 541 mL + 432 mL = 973 mL

64) 717 mL + 199 mL = 916 mL

65) 374 mL + 262 mL = 636 mL

66) 679 L + 226 L = 905 L

67) 579 mL − 401 mL = 178 mL

68) 705 g + 33 g = 738 g

**Part 8** Remember? Problems from earlier packets.

69) 6 × 8 = 48 so 6 × 4 = 24

70) 4 m by 7 m: area = 28 sq m

71) 25 minutes after 9:50 = 10:15

72) Round 638 to the nearest ten: 640

## Mass and Liquid Volume: Practice (continued)

**Part 8** Remember? (continued)

73) $81 \div 9 =$ **9**

74) $306 + 495 =$ **801**

**Part 9** Word problems Draw a bar if it helps.

75) A bag of apples weighs 3 kg. How much do 4 bags weigh?

Number sentence **4 × 3 = 12**

Answer **12** kg

Draw or show your work

76) A pitcher holds 2 liters. How many pitchers fill an 18-liter tub?

Number sentence **18 ÷ 2 = 9**

Answer **9** pitchers

Draw or show your work

77) Mia's backpack weighs 4 kg. Her lunch adds 1 kg and a book adds 2 kg. What is the total?

Number sentence **4 + 1 + 2 = 7**

Answer **7** kg

Draw or show your work

## Mass and Liquid Volume: Practice (continued)

### Part 9 Word problems (continued)

78) A fish tank holds 40 liters. It has 23 liters in it now. How much more water will fill it?

Number sentence **23 + 17 = 40**

Answer **17** liters

Draw or show your work

79) Challenge: A box holds 6 bags of flour. Each bag is 2 kg. The empty box is 1 kg. How heavy is the full box?

Number sentence **6 × 2 = 12, 12 + 1 = 13**

Answer **13** kg

Draw or show your work

# Check-In: Mass and Liquid Volume

Show what you know. No help this time.

**Mastery: 5 of 6 (80%)**

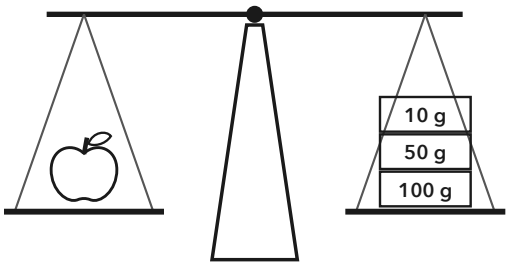
**Part 1** Grams or kilograms?

1) **A coin is about 3** (grams) kilograms

2) **An eraser is about 20** (grams) kilograms

**Part 2** Read the balance

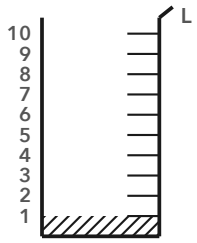
3)




100 + 50 + 10 = 160 g

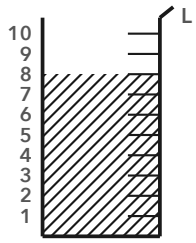
**Part 3** Read the beaker

4)



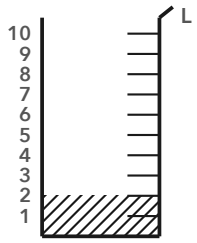

1 liters

5)




8 liters

6)




2 liters